MIL-E-47231(MI)
26 July 1974
Superseding
MIS-15748
14 December 1965

MILITARY SPECIFICATION

ELECTRICAL COMPONENTS, POTTING OF

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements of a polysulfide base compound for potting or sealing electrical components.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-S-8516	Sealing Compound, Polysulfide Rubber, Electric Connectors and Electric Systems, Chemically Cured

3. REQUIREMENTS

3.1 Material. Polysulfide compound and accelerator in accordance with MIL-S-8516, Class 2.

3.2 Apparatus.

a. Scale or balance accurate to 0.5 percent.

b. Pneumatic injector gun with tip suitable for the even flow of viscous fluids.

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c. Deep freeze capable of maintaining minus 10 degrees Fahrenheit (F) to minus 50 degrees F.

3.3 Procedure. The sealant specified is a two part material consisting of a base compound and an accelerator of contrasting colors. The two compounds are mixed prior to use.

3.3.1 Pre-mixed sealant.

3.3.1.1 The potting compound may be received pre-mixed and packed in dry ice. The cartridges containing the pre-mix shall be stored in a deep freeze between minus 10 to minus 50 degrees F. Storage at this temperature extends the use life up to 20 calendar days.

3.3.1.2 Remove the required numbers of pre-mix tubes from the deep freeze and allow to return to room temperature (usually 30 to 60 minutes). The use life of the pre-mix after reaching room temperature is approximately 90 minutes. Discard any partially used pre-mix tubes. Do not replace in deep freeze for future use.

3.3.2 Sealant not pre-mixed.

3.3.2.1 The material may be received in bulk or in pre-weighed unit packages. If received in bulk, the two components must be weighed and mixed according to vendor's instructions.

3.4 Potting procedure.

3.4.1 Inspection. All connectors, terminations and electrical components must be electrically inspected before potting.

3.4.2 Cleaning. All components must be thoroughly cleaned before potting. Vapor degrease with trichlorethylene, or equivalent, or wipe surfaces with acetone, methyl ethyl hetone, or equivalent.

3.4.3 Coating. All surfaces, such as mold surfaces, where adhesion of the sealant is not desired must be coated with a mold release compound. A suitable silicone or silane is recommended.

3.4.4 Filling. Using a pneumatic injector gun or other suitable tool, fill the mold with the mixed potting compound. Fill from bottom to top and from the center outward being careful not to include air bubbles. This is best done by placing the tip on the center-bottom and drawing it up slowly as the void fills with sealant.

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3.4.5 Air bubbles. Air bubbles which rise to the surface can be removed by pricking with a a. stiff wire or sharp object.

3.4.6 Cure. Cure at room temperature for 24 hours minimum or at 125 degrees plus or minus 5 degrees F, for one hour minimum.

4. QUALITY ASSURANCE

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Finished product. The finished product shall be reasonably free of voids and of such quality as to prevent dirt and moisture from reaching soldered joints. The finished product shall meet all the contours and requirements of the applicable engineering drawings.

5. PREPARATION FOR DELIVERY

This section not applicable to this specification.

6. NOTES

6.1 Intended use. This specification covers the requirements of a polysulfide base compound for potting or sealing electrical components.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-15748, dated 14 December 1965.

Custodian:
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